

Predicting Depression Type Using Lifestyle and Behavioral Factors: A Multi-Dimensional Analysis

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Project element	Final specification
Dataset	1,998 observations, 21 variables, 12 target classes
Data quality	0 missing values and 0 duplicate rows
Models compared	Standardized Logistic Regression and Gaussian Naive Bayes
Primary evaluation metric	Macro F1-score because the target is multi-class and imbalanced
Companion artifact	Notebook included; all figures in this report were generated from code

This report evaluates depression type classification using the full feature set of the uploaded dataset. The document combines exploratory data analysis, statistical reasoning, model comparison, and interpretation of variable importance. The main objective is not only to identify which model performs better, but also to understand why one approach is better aligned with the structure of the data.

1. Introduction

Depression is a complex mental health condition influenced by psychological, behavioral, and social factors. Rather than being driven by a single variable, it reflects interactions across multiple dimensions.

This study aims to classify Depression_Type using a multi-dimensional approach based on the full set of predictors. Two models are compared: Logistic Regression and Naive Bayes, which differ in how they handle relationships between variables.

Research objectives

- Explore relationships between all available predictors and the target variable Depression_Type.
- Use appropriate statistical methods for categorical and numerical variables rather than applying a one-size-fits-all approach.
- Evaluate multi-class classification performance using both accuracy and Macro F1-score.
- Interpret why Logistic Regression and Naive Bayes differ in performance on this dataset.

2. Dataset Description

The dataset used in this study was obtained from Mendeley Data (Choudhury, 2022).

The dataset contains 1,998 observations and 21 variables, including the target variable Depression_Type. All variables are integer-coded, but they represent different types of information and should not be interpreted uniformly.

Predictors include demographic, psychological, behavioral, and support-related factors. Psychological variables represent internal mental states, while behavioral and contextual variables provide additional information about daily patterns and environment.

A data quality check confirms that there are no missing values or duplicate records. The target variable is multi-class and imbalanced, making balanced evaluation metrics necessary.

To guide the analysis, variables are classified based on their statistical roles rather than their raw data types.

Predictor Variable Classification

Variable Type	Variables	Interpretation
Numerical (continuous-like)	Age, Sleep_Hours, SocialMedia_Hours	Represent measurable quantities where differences and averages are meaningful
Ordinal (ordered categories)	Low_Energy, Low_SelfEsteem, Nervous_Level, Your overeating level, Depression_Score	Represent ordered levels (e.g., low to high), where ranking matters but spacing between values may not be equal

Categorical (nominal)	Gender, Education_Level, Employment_Status, Symptoms, Search_Depression_Online, Worsening_Depression, How many times you eat, SocialMedia_WhileEating, Coping_Methods, Self_Harm, Mental_Health_Support, Suicide_Attempts	Represent distinct groups with no inherent order
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Although all variables in the dataset are integer-coded, they do not all represent true numerical quantities. Some variables function as categorical or ordinal indicators rather than continuous measures.

In particular, variables such as Low_Energy, Low_SelfEsteem, and Overeating_Level are best interpreted as ordinal, as they represent ordered levels rather than exact numeric values.

For the purpose of analysis, ordinal variables are treated as numerical to preserve their ordering and simplify modeling. This is a common approach in practice, although it assumes equal spacing between levels, which may not strictly hold.

3. Exploratory Data Analysis

The target variable **Depression_Type** is encoded numerically in the dataset. Each value represents a specific category of depression. For clarity and interpretability, the mapping between encoded values and their corresponding clinical descriptions is provided in the table below.

Code	Type	Code	Type
0	No clinically significant	6	Seasonal affective
1	Mild depression	7	Postpartum depression
2	Moderate depression	8	Bipolar-related episode
3	Moderately-severe	9	Reactive depression
4	Severe depression	10	Psychotic depression
5	Dysthymia	11	Other specified

This encoding is used consistently throughout the analysis to interpret model results and variable relationships.

Based on this classification, different statistical methods are applied to numerical and categorical variables in the following analysis.

3.1 Class Distribution

The distribution of **Depression_Type** is imbalanced, with some categories appearing much more frequently than others (see Appendix A, Figure A1). This is important because a model could achieve high accuracy by favoring majority classes while performing poorly on minority ones.

For this reason, accuracy alone is not sufficient. The analysis later relies on **Macro F1-score**, which evaluates performance across all classes more fairly.

3.2 Correlation Matrix

A correlation overview shows that several predictors—especially psychological and behavioral variables—are moderately related (see Appendix A, Figure A2). This indicates that the dataset contains overlapping information rather than independent signals.

This is relevant for model selection: **Naive Bayes may be limited by this dependency**, while **Logistic Regression is better suited to handle correlated features**.

3.3 Categorical Variables and Chi-Square Analysis

Categorical variables are analyzed using the Chi-Square test of independence, since their relationships with Depression_Type cannot be captured using correlations or averages.

The null hypothesis (H_0) assumes independence between the variable and Depression_Type, while the alternative hypothesis (H_1) assumes an association.

To formally test these relationships, a **Chi-Square test of independence** is applied. The results are summarized in the table below. All variables show extremely small p-values, leading to the rejection of the null hypothesis of independence.

Variable	Chi-square	p-value	df
Symptoms	10409.47	<1e-300	154
Coping_Methods	6327.15	<1e-300	143
Employment_Status	1902.06	<1e-300	44
Education_Level	1141.59	1.34E-218	33
SocialMedia_WhileEating	754.93	2.06E-137	33
Mental_Health_Support	421.00	2.12E-83	11
Search_Depression_Online	352.40	7.52E-69	11

Worsening_Depression	298.91	1.48E-57	11	
Self_Harm	233.77	6.94E-44	11	
Suicide_Attempts	46.03	0.065	33	
Gender	10.24	0.509	11	
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This indicates that categorical variables are significantly associated with **Depression_Type**. In particular, **Coping_Methods** and **Employment_Status** display strong patterns in both the visualizations and statistical results, suggesting that they are important predictors.

Detailed visualizations of categorical variables are provided in the Appendix (Figures A3–A5). These plots show that variables such as education level, employment status, and coping methods are unevenly distributed across depression types.

These patterns suggest that categorical variables are associated with the target and contain meaningful information for classification.

3.4 Numerical Variables

Numerical and ordinal variables are analyzed to assess how well they differentiate between depression types. As shown in Appendix A (Figures A6–A7)

In particular, **Low_Energy** varies significantly across depression types, indicating strong predictive potential. In contrast, **Sleep_Hours** and **SocialMedia_Hours** show overlapping distributions, suggesting weaker discriminative power.

Statistical results further support these observations. Variables such as **Low_Energy** show significant differences across groups, while others do not.

3.5 Key Insights

- The dataset is **class-imbalanced** (Appendix A, Figure A1)
- Predictors are **not independent** (Appendix A, Figure A2)
- **Categorical variables show strong associations** (Appendix A, Figures A3–A5)
- **Low_Energy is the strongest numerical predictor** (Appendix A, Figure A6)
- Some intuitive variables (e.g., sleep_hours) are **less informative than expected**

All supporting visualizations are provided in the **Appendix**.

4. Statistical Reasoning and Substantive Interpretation

The EDA results reveal a multi-dimensional data structure. Psychological variables such as Symptoms, Low_Energy, Low_SelfEsteem, Nervous_Level, and Depression_Score are conceptually close to the target because they reflect internal mental states. By contrast, variables such as Sleep_Hours or

SocialMedia_Hours provide contextual information, but usually with weaker class separation when considered alone.

This difference does not mean the contextual variables are irrelevant. Their value may lie in how they interact with more direct indicators. For example, social media behavior, support access, and coping choices may not be dominant predictors in isolation, yet they can enrich the model by providing complementary context to the psychological features. That is exactly why a full-feature modeling approach is preferable here.

The statistical reasoning also links directly to model assumptions. The Chi-Square results show that several categorical variables are strongly associated with Depression_Type. The correlation matrix shows that many predictors are related rather than independent. The numerical analysis shows that some variables carry strong signal while others are more weakly informative. Taken together, these results imply that the dataset contains overlapping and interdependent information, which makes Naive Bayes a conceptually weaker fit than Logistic Regression.

5. Modeling Methodology

To preserve the full structure of the dataset, all available predictors are included in the baseline models. The data is split into training and testing sets using an 80/20 stratified split, ensuring that each class is proportionally represented in both subsets. This is especially important in an imbalanced multi-class problem, as it helps maintain consistency between training and evaluation.

Two models are used in this study. The first is Logistic Regression with standardization applied through a pipeline. The second is Gaussian Naive Bayes, which serves as a contrasting baseline model.

Logistic Regression is chosen because it can handle situations where variables provide overlapping or related information. It also offers a level of interpretability, making it easier to understand how different features contribute to predictions. In contrast, Naive Bayes assumes that all predictors are independent, which makes it simpler but potentially less flexible when variables are related.

Model performance is evaluated using both accuracy and Macro F1-score. While accuracy provides a general overview, Macro F1-score is more appropriate in this context because it treats all classes equally and gives a clearer picture of performance across the full set of depression types.

6. Model Results

Model	Accuracy	Macro F1	Interpretation
Logistic Regression	0.7175	0.8659	Best overall balance; strong fit to the dataset structure
Naive Bayes	0.5325	0.7563	Lower performance; likely constrained by independence assumption

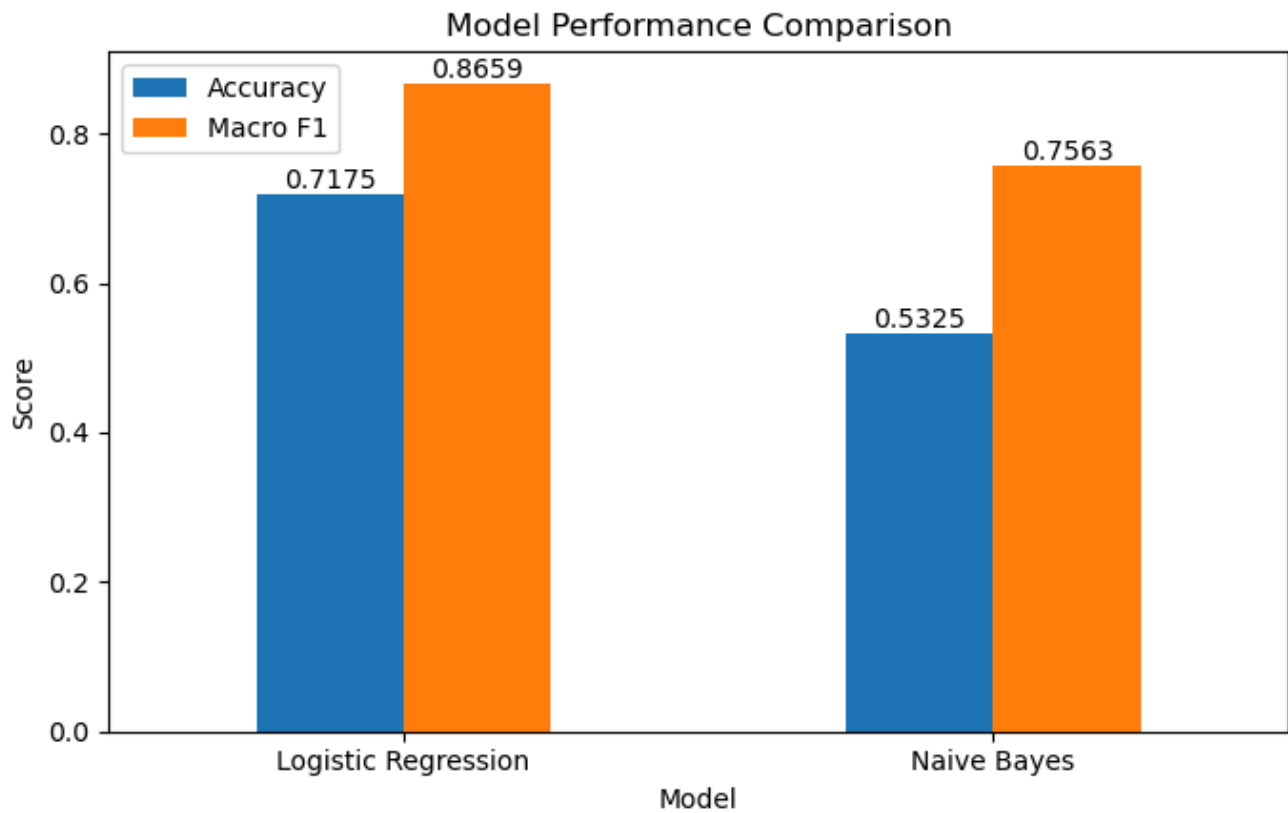


Figure 1. Accuracy and Macro F1-score for Logistic Regression and Naive Bayes.

The results show a clear difference between the two models. Logistic Regression achieves higher accuracy and a higher Macro F1-score compared to Naive Bayes.

More importantly, the improvement is not limited to the majority classes. The higher Macro F1-score indicates that Logistic Regression performs better across all depression types, including the less frequent ones.

This result is both practical and conceptual. From a practical perspective, it means that Logistic Regression provides more reliable predictions. From a conceptual perspective, it suggests that the dataset does not behave like a collection of independent variables.

Instead, the predictors appear to contain overlapping and interacting information. Because Logistic Regression can account for these relationships, it is better suited for this dataset than Naive Bayes.

6.1 Interpretable Feature Signals

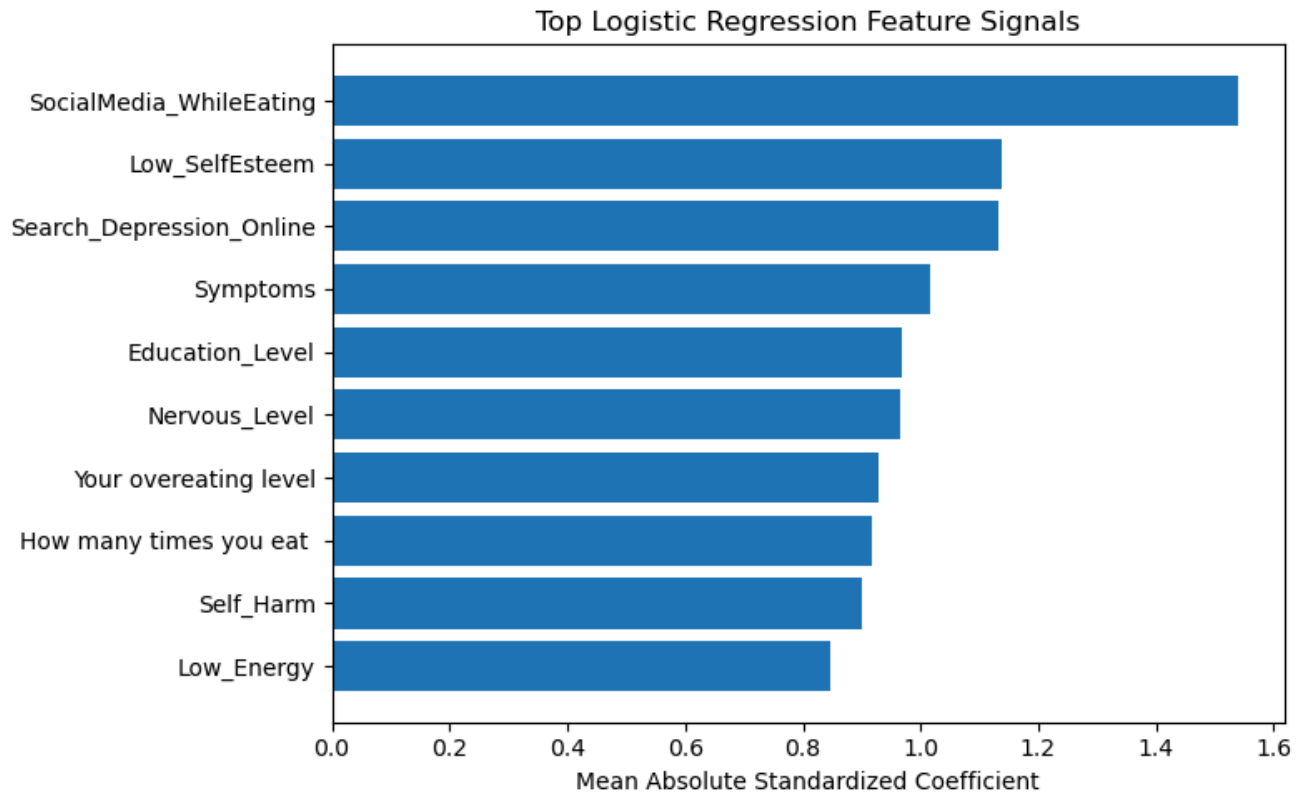


Figure 2. Ten strongest average absolute standardized coefficients from Logistic Regression.

To better understand how the model makes predictions, feature importance is examined using the standardized coefficients from Logistic Regression.

The results show that variables such as SocialMedia_WhileEating, Low_SelfEsteem, Search_Depression_Online, Symptoms, Education_Level, and Nervous_Level are among the most influential predictors.

This pattern reinforces an important insight: depression classification is not driven by a single type of variable. Instead, both psychological indicators and contextual factors contribute to the model's performance.

It is important to note that feature importance in Logistic Regression does not imply causation. However, it does provide a useful way to identify which variables the model relies on most when making predictions.

6.2 Diagnostic View

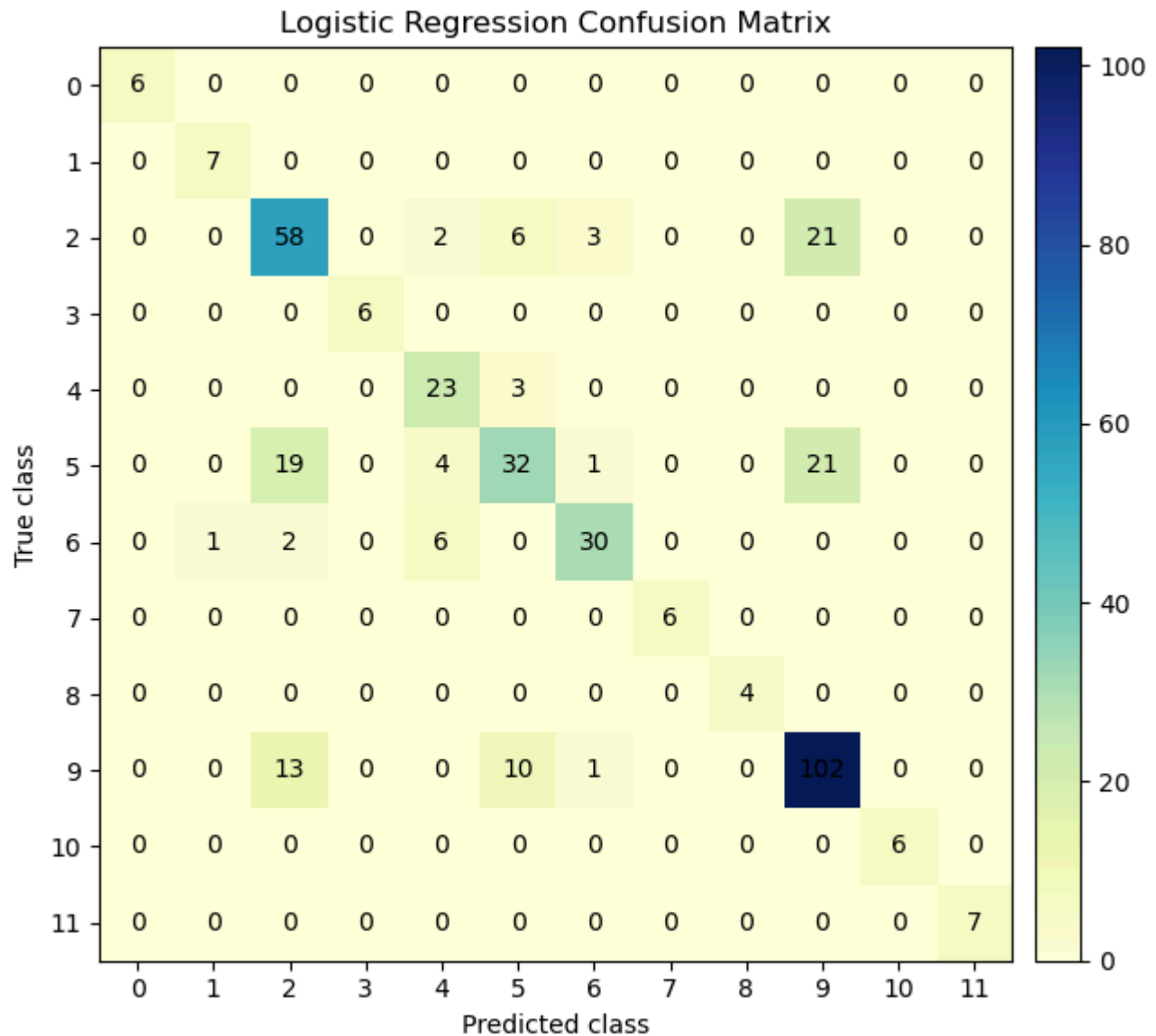


Figure 3. Confusion matrix for Logistic Regression on the holdout test set.

The confusion matrix provides a more detailed view of model performance across all classes.

Most predictions fall along the diagonal, which indicates that the model is able to correctly classify a large portion of the observations. However, there are still some off-diagonal errors, meaning that certain classes are occasionally confused with one another.

This is expected in a multi-class mental health problem, where some depression types may share similar characteristics. As a result, even a strong model will not achieve perfect separation between all classes.

Overall, the confusion matrix suggests that the model performs well, while also highlighting the inherent complexity of the task.

7. Discussion

The results suggest that depression classification is inherently multi-dimensional. No single variable is sufficient on its own, and meaningful prediction emerges from combining psychological, behavioral, and contextual factors.

Psychological variables, such as Low_Energy and Symptoms, provide the strongest direct signals. In contrast, behavioral variables such as sleep duration show weaker individual effects, highlighting the importance of data-driven evaluation rather than intuition.

The comparison between Logistic Regression and Naive Bayes further supports this conclusion. Logistic Regression performs better because it can account for relationships between variables, while Naive Bayes is limited by its independence assumption.

Overall, the findings indicate that depression is influenced by interacting factors, and models that capture these relationships are better suited for this task.

8. Limitations and Future Work

Although the results are strong, several limitations should be considered.

First, the dataset may not capture all aspects of mental health. Depression is influenced by many factors, some of which may not be included in the data.

Second, future work could extend this analysis in several ways. More advanced models could be tested to determine whether additional complexity improves performance. Class-level evaluation could also be explored in more detail to better understand which depression types are more difficult to predict.

Additionally, feature engineering techniques could be used to improve both interpretability and performance.

9. Conclusion

This study shows that depression type classification requires a data-driven and multi-dimensional approach.

The dataset contains meaningful structure across psychological, behavioral, demographic, and support-related variables. Both exploratory analysis and statistical testing suggest that these variables are not independent, but instead interact with one another.

Logistic Regression outperforms Naive Bayes, achieving higher accuracy and Macro F1-score. This result is consistent with the earlier findings and supports the idea that models capable of handling overlapping information are better suited for this problem.

Overall, the results highlight that psychological variables provide strong direct signals, while contextual variables enhance prediction when used together. This combination leads to a more complete and realistic understanding of depression classification.

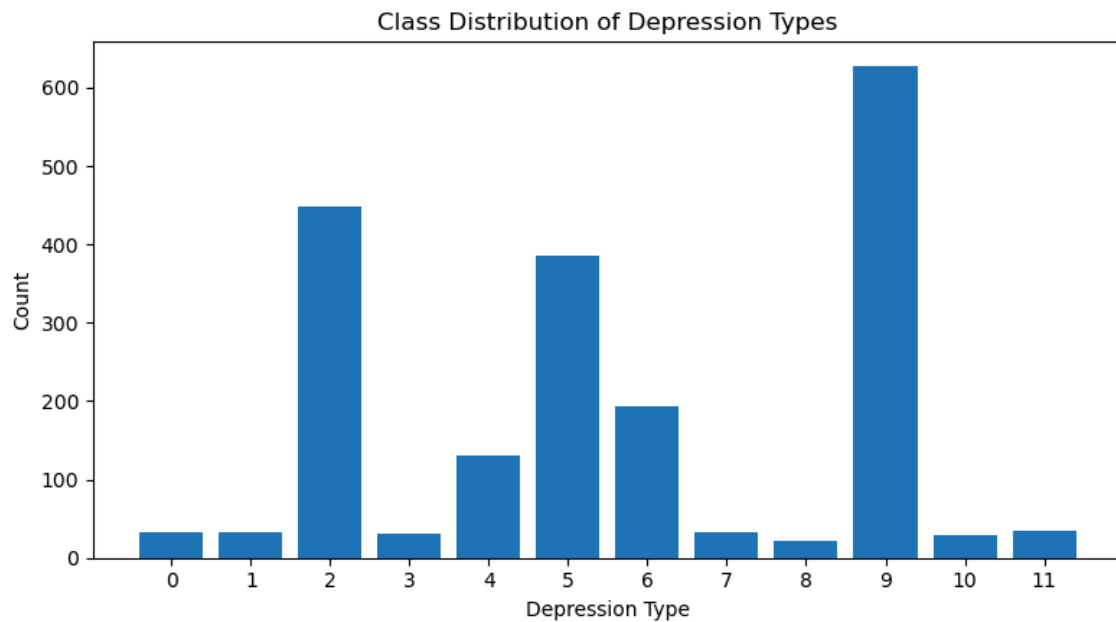
References

Choudhury, A. (2022). *Mental health dataset* [Data set]. Mendeley Data.
<https://data.mendeley.com/datasets/xppzm3kv9g/2>

Appendix A: Additional Visualizations

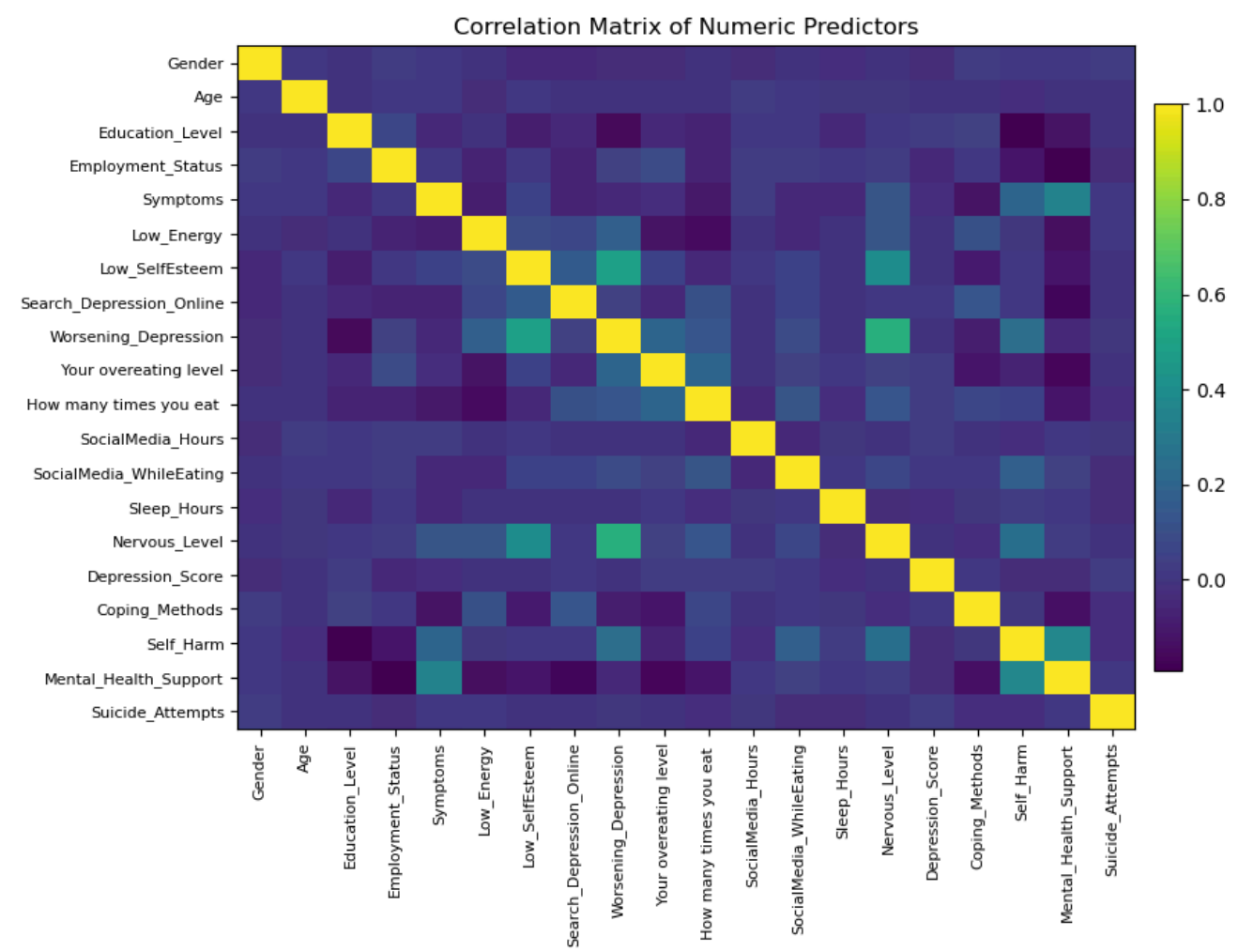
This appendix contains supplementary figures that support the analysis presented in Section 3. These visualizations provide additional detail but are not essential for understanding the main conclusions.

Figure A1: Class distribution of Depression_Type



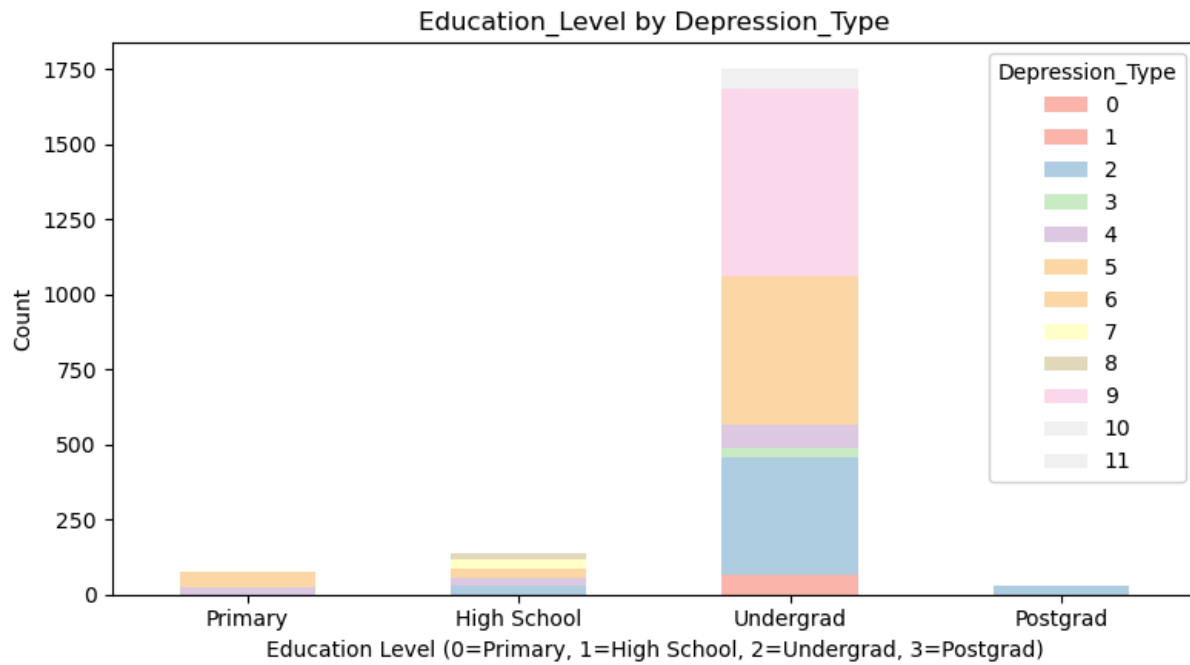
This figure shows the frequency of each depression category in the dataset. The distribution is clearly imbalanced, with certain classes dominating the sample, which may bias model performance if not properly addressed.

Figure A2: Correlation matrix of predictors



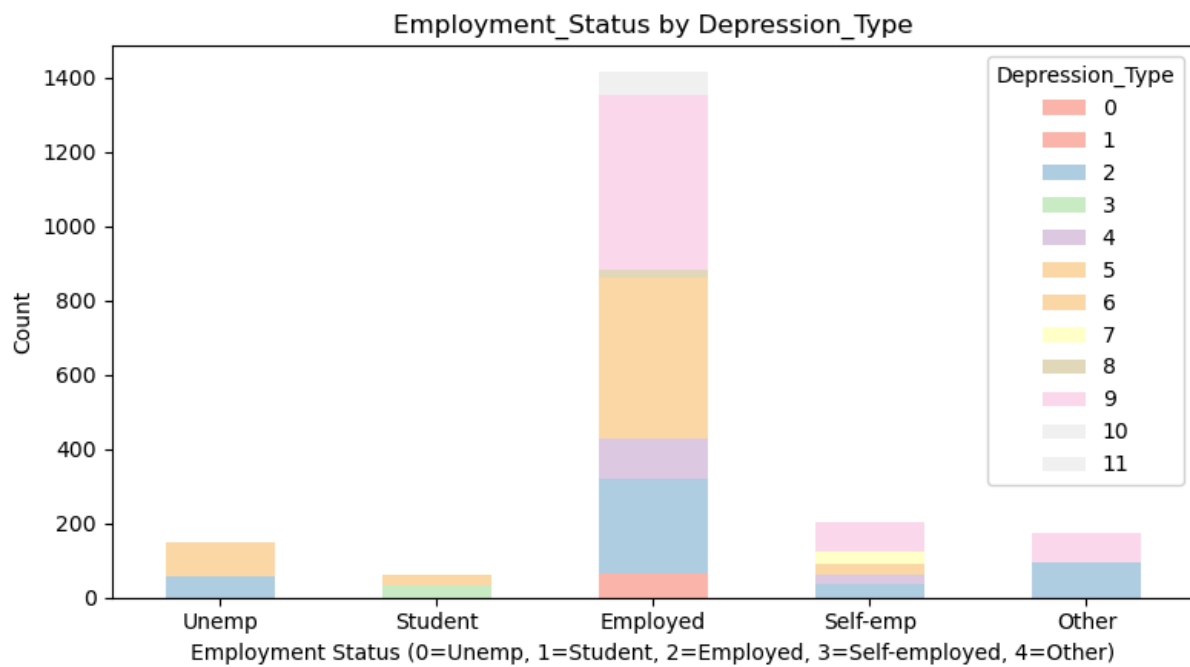
This heatmap illustrates the pairwise correlations between predictors. Several psychological and behavioral variables show moderate correlation, indicating that features are not independent and may contain overlapping information.

Figure A3: Education level vs Depression_Type



This plot displays the distribution of education levels for each depression category. The uneven patterns suggest a potential relationship between education and depression type.

Figure A4: Employment status vs Depression_Type



This figure shows how employment status varies across depression categories. Certain employment groups are more concentrated in specific depression types, indicating a possible socioeconomic influence.

Stacked bar chart titled "Coping_Methods by Depression_Type". The Y-axis represents the "Count" (0 to 400). The X-axis represents "Coping_Methods" (0 to 13). The legend lists 12 Depression_Type categories (0 to 11) with corresponding colors. The chart shows the distribution of coping methods across different depression types.

Coping_Methods	0	1	2	3	4	5	6	7	8	9	10	11
0	0	0	135	0	0	0	0	0	0	0	0	0
1	0	25	0	0	0	0	0	0	0	0	0	0
2	0	0	0	30	0	170	0	0	0	115	0	0
3	0	0	55	0	30	0	0	0	0	90	0	0
4	0	0	0	0	0	0	35	0	0	0	0	0
5	0	0	0	0	0	55	0	0	0	0	0	0
6	0	30	0	0	0	30	0	0	0	0	0	0
7	0	0	0	0	20	25	0	30	0	0	0	0
8	0	0	35	0	0	0	0	0	0	0	0	0
9	30	0	125	0	75	55	0	0	0	165	0	0
10	0	0	0	0	0	25	0	0	0	50	0	0
11	0	0	70	0	0	60	0	0	0	105	0	0
12	0	0	0	0	0	0	0	0	15	0	0	0
13	0	0	15	0	0	25	0	0	0	25	0	0

Figure A6: Low_Energy vs Depression_Type

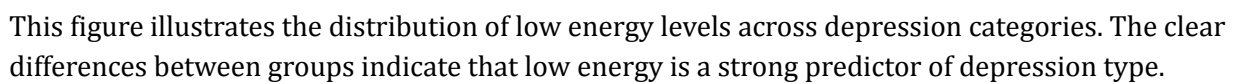
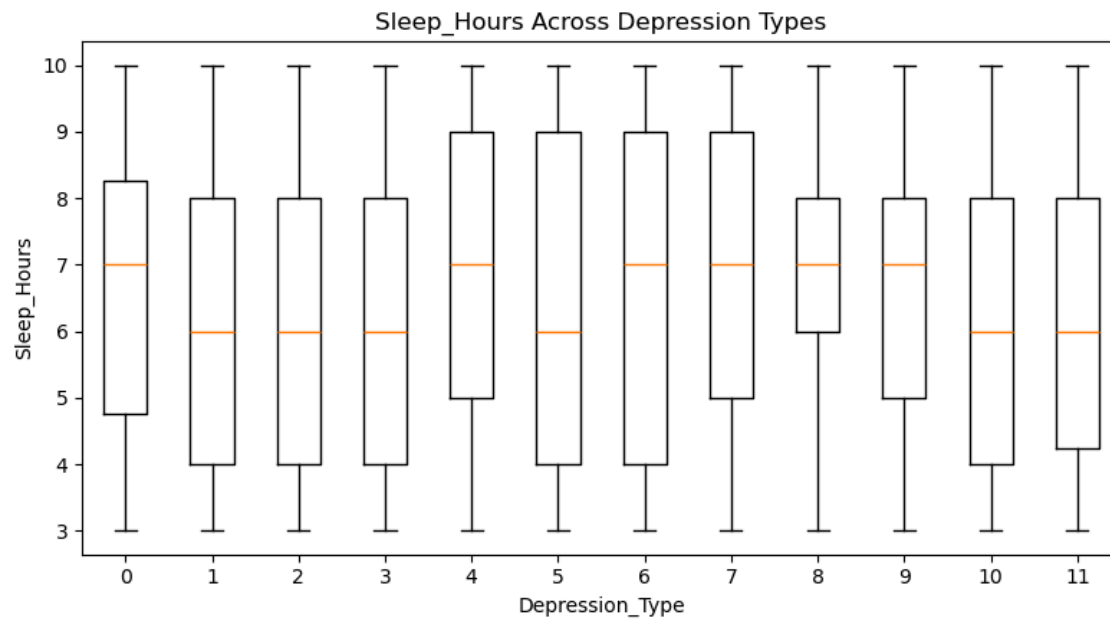


Figure A7: Sleep_Hours vs Depression_Type



This plot shows the distribution of sleep duration across depression categories. The substantial overlap between groups suggests that sleep alone is not a strong distinguishing factor.